DATA PROGRAMMING PROJECT REPORT

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Introduction

Analysing the trends of property supply and prices is something key, in many aspects of society, such as the government and estate agents. Being able to see what shapes the supply, demand and prices of housing can help with future predications and therefore, planning.

The goal of this project is to use Python Data Programming techniques to explore, clean and analyse data, as well as connecting that data to a database, using a variety of Python available libraries.

The dataset I have chosen has more than enough records for this purpose and it will be looking at the trend of property prices across the country of Ireland in 2025.

This project satisfies the learning objectives from the project brief.

Project Objectives

The objectives of this project is to locate one or more suitable datasets (with permission to use those one or more datasets), they must also be comprised of at least 1000 records/rows and perform analyses of them using Python and its associated libraries (Pandas, NumPy, matplotlib, SciPy etc).

The data must be pre-processed and transformed. This involves cleaning operations, such as checking for duplicate values or null values and cleaning them from the dataset, and joining one or more datasets, where appropriate (they must be related in some way).

The processed data must then by stored in a database of choice for retrieval.

The data must then by analysed and visualised, providing 4 insights into the data.

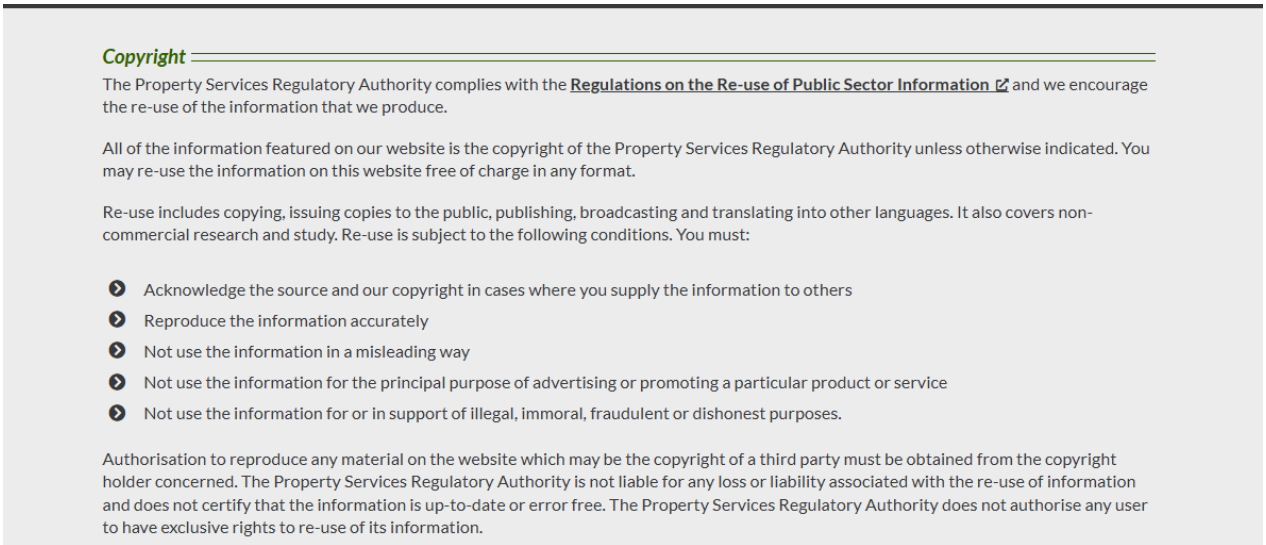
A project report must be written, as well as a video presentation detailing the project.

Dataset Description

Dataset Chosen: House prices for all months in 2025 across all countries

Dataset Link: propertypricerregister.ie (2025) *Residential Property Price Register* Available at: https://propertypriceregister.ie/website/npsra/pprweb.nsf/PPRDownloads?OpenForm&File=PPR-2025.csv&County=ALL&Year=2025&Month=ALL [Accessed 14th December 2025]

Permission to use dataset:



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I chose a dataset about the property prices in all the counties in Ireland, for every month in the year 2025 because I thought this would be a good dataset filled with valuable data to analyse and see trends, numerical data, relationships, patterns, and correlations. It is a publicly available dataset for the Property Services Regulatory Authority, with permission to re-use.

From briefly analysing my dataset using *df.head()* and *df.info()*, I discovered my dataset is a mixture of categorical and numerical variables and that there are 53,301 records/rows and 9 columns. There is missing data in the Eircode column and there is an entirely empty column (Property Size Description). The data types are float64 (1) and object (8), so mostly text (as is common for CSV files).

Pre Processing Tasks

- ***Exploring my data (exploration.py):*** loading in my downloaded data and creating a data frame. Use *df.head()* to get information about the first few records/rows (and print that) and then use *df.info()* to get more general information about my dataset. Columns that could be useful for analyse could be: County & Price. Explored the top 5 countries, using *.values_counts():* Dublin, Cork, Kildare, Galway, Meath. There are 13,387 null values in Eircode. Also, through using *.duplicated.sum(),* discovered there is 52 duplicates. Using *.describe()* for pricing, I discovered the most common price is 300,000 and that appears 674 times. This column is still an object, so this will need changing to numeric during cleaning. Although the price variable was converted to a numeric data type (*to_numeric()*), values came back as NaN, this needs furthering preprocessing at the next stage. The datetime appears to be successful (*to_datetime)*.
- ***Data Cleaning & Transformation (cleaning.py):*** I used *.drop()* to get rid of the Property Size Column, as it was completely null and should be removed, as it cannot offer me anything to analyse. I then converted Date of Sale to datetime, which was successful *(to_datetime*), the prior data type that it was in was a String, so I wanted Python to realise that data was a time. I started working on converting the Price column next, as that was also a string (text). I had several issues here, particularly with the euro sign and the spacing. I initially tried to use *rename* to rename the column to something safer and cleaner but was constantly getting errors when printing. So, I used Python list comprehension and used a for loop for iterate and see if anything started with "Price_," for renaming. That was then successful. I then converted it to a numeric value, so that Python knows I am referring to a number within my data. I then used *.dropna* to get rid of any missing prices and used the < = operators to get rid of any values that were less than or equal to 0, because obviously a house would not be sold for that amount, so that is a data issue. I also removed any properties that sold for below market price because these may not be the best for data analyse, as they could

indicate bias or family discount. I then created two new columns from “Date_Of_Sale,” which were “year” and “Month,” this is for better analysis, as house prices fluctuate monthly, yearly, and that is what I want to focus on. I then used *.info, .describe and .sum* to check my data frame information, the information for my Price_EUR column (which I had altered) and checking for any Num values, to make sure I had gotten rid of them. I then saved my clean data to a csv file called “ppr_clean.csv,” using *to_csv*.

Database Storage

I chose MySQL Workbench as my database management system of choice, as I have used this in a couple of different modules before and am familiar with it.

I created a *‘property_data.sql’* file and then created a *property_data* database, which I then highlighted to use. I then used a CREATE TABLE statement to create a basic table skeleton for my processed data. I called the table *ppr_clean*.

I then created a database file on VSCode, with Python. I loaded in my processed data in the file and then created my SQL connection using *.connect()* and *.cursor()* (for executions). I created an INSERT statement, making sure that the columns matched the order in MySQL Workbench. I also masked the NaN values in my data, as “None,” so that they would be recognized by MySQL Workbench. I used *.executemany()* and *.commit()* to run my query and commit it and then finally used a *.close()* to shut down the connection and the cursor.

Analysis

Insight 1

How have the prices of properties changed over time in Ireland?

I used .*groupby()* on the Month and the Price_EUR columns and also .*median()* to be able to see more of a difference in the prices throughout the year, as median is not as affected by outliers.

I used .*plot*() to create a line graph because it is useful for viewing trends.



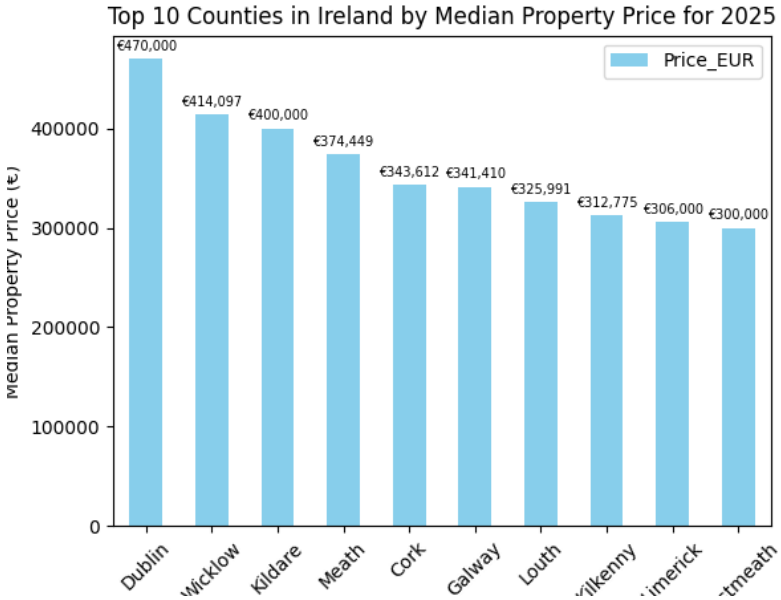
As you can see from this graph, the median price for properties starts at slightly less than 345,000 in January and peaks at 365,000 in August before dropping again and stabilising somewhat, which suggests a high demand, not enough supply of properties across the country in the month of August, which is also just before the start of the school year, which could be a factor. February is the lowest at 335,000, which suggests there were more properties on the market in this month than at any other month in the year or that there was less demand, perhaps because it was not long after the holiday season. The median property price started climbing in April.

Insight 2

How have prices fared across different counties in Ireland in 2025?

I used .*groupby()* and .*median()* on the County and Price_EUR columns, to find the median prices across the counties in Ireland, for comparison. I decide to use the top 10, so the visualisation would not look too busy and overwhelming.

I used a bar chart (.*bar*) for this as it is the best way to compare values across various categories.



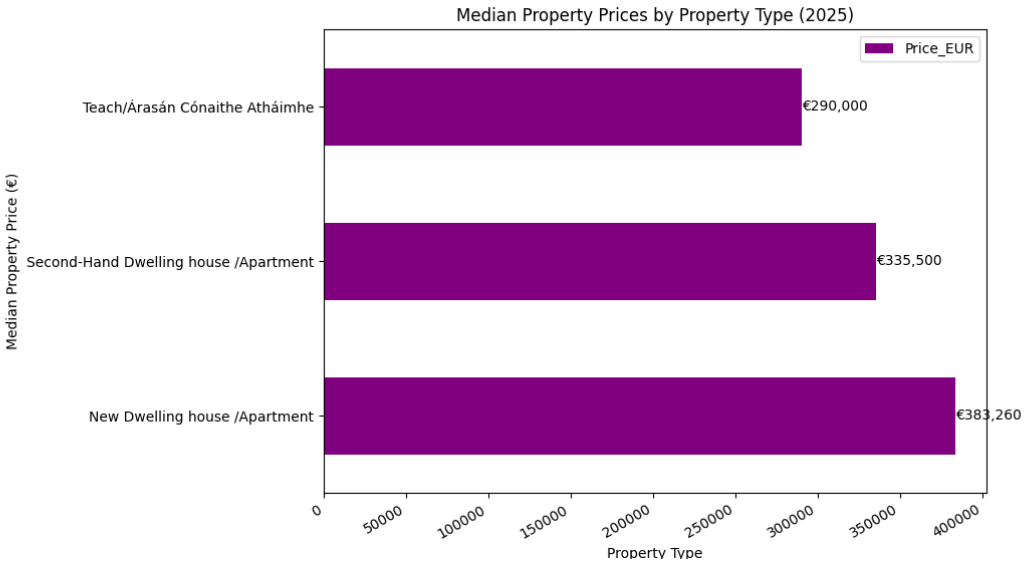
As you can see from this graph, property prices amongst the 4th through 10th counties are all in the same range (300,000s), these are the less urban and less populated areas, whereas the first 3 are in the 400,000s, these are busier and more urban areas. Dublin has the highest median price, given that it is the capital of Ireland and a major city, for work, education, and tourism, as expected. Housing supply is also limited because of this, pushing up the median price. Wicklow and Kildare, in the top 3, are often considered great commuter counties to get to Dublin, pushing up the prices in those areas. This graph highlights some regional inequality within the Irish housing market. Location and access to employment, education and tourist attractions plays a big part in determining the housing prices.

Insight 3

Are there price differences between newer built properties and older properties?

The method I used for this is a horizontal bar chart (*.barh*) as I want to compare the values of older properties and newer properties and their prices, to see if there's a difference and a horizontal would give me more room for what I believed were going to be long label names.

I used *.groupby()* and *.median()* once again on the Description_Of_Property and Price_EUR columns. Median is less affected by outliers, than the mean.



This graph shows the median prices for the year 2025 across the different counties of Ireland, for uninhabited properties, newly build properties and older properties.

The price difference is not as wide as I expected it to be. But as expected, newly built properties are more expensive than second-hand/older properties and uninhabited ones.

Uninhabited properties are still commanding the price of their locations, but the work necessary to bring them up to a liveable standard means they are cheaper than the other options.

The second-hand dwellings/older properties are 2nd here, according to the chart, offering a more affordable option for buyers in the market. This would be down to cheaper building costs at the time of construction (such as materials). Also, these older properties may not meet the requirements for modern building standards and energy efficiency standards.

Modern/Newly built homes are the most expensive, in median price, as expected, due to the rising costs for construction. This is down to factors such as a shortage of labour and the rising costs of construction materials. Modern buildings now also have a wider range of requirement they must meet, regarding building standards and energy requirements, thus driving up cost.

Insight 4

Are there any outliers that have affected the median property prices across Ireland?

I used a box plot (*.box*) for this analysis, as a box plot helps to visualise the spread, median, quartiles and outliers of any given data, which is what I wanted to see here.



This boxplot reveals a skewed distribution of property prices in 2025 across the different counties of Ireland, which is not entirely unexpected, as there usually cases of property outliers. While many of the sales of properties this year are clustered together around a comparable price range (about the 300,000 mark), several high-value outliers are present in the boxplot. These extreme values/outliers drastically exceed the normal price range (median) for property prices across the different counties in Ireland and represent luxury properties or properties in much desired locations, such as cities or urban areas (Dublin for example). This also confirms about why I decided to use median in my earlier analyses, as these outliers would have skewed my analyse if I had used mean.

Critical Discussion and Challenges

A challenge I faced during this project was to find a good dataset with over 1000 records/rows and permission to use it. It was difficult to also find a dataset that was not already frequently modelled online.

Another challenge I came across was dealing with the conversion of the Price column from a string (text) that it was in, into a number, so that Python can deal with it as a number. I tried to rename the column, at first, but it would not convert to numeric. So, I ended up using list comprehension and a loop, searching for any column that featured: “Price_” and then renaming the column based on any value that started with it. I figured it was an issue with extra spaces, characters or encoding that I could not see, but was causing issues.

Placing my processed data in MySQL Workbench gave me a considerable number of issues. I tried using MySQL Workbench at first, and using the import Wizard, which I had encoding issues to start off with (UTF-8, Latin1), and when I fixed those, records still did not seem to be loading into my table.

I then went over to VSCode to try and do it that way using Python, where I also suffered the same problems of getting 0 records when I tried to query to test it on MySQL Workbench. That triggered me to go back to my **cleaning.py** file and see if I had done something wrong. I ended up having to remove the filtering I did with prices below full market price, as it was getting rid of all my records altogether.

I did not realise how easy it is to suffer data loss.

I also had issues with the price conversion section, because of the way my data (original CSV file) is structured, I had to ensure, using regular expressions, that what I was looking for was not a digit or a dot. This was to take away any hidden characters in the code, which caused conversion issues, as my original data had a euro sign and underscores.

I made sure all the columns were in the correct order, so MySQL Workbench would not run any errors, as that was another issue I encountered. I also ended up having to use a mask because of my NaN values. MySQL Workbench was not recognizing them as being NULL values.